N_TOF EAR2 · ^{3}HE(N,F) → X17 / IPC · THERMAL WINDOW

MX17 / X17 — progress summary

Beam commissioning, DAQ, and trigger design

Work of the weekend + Monday morning — 18–20 July 2026

Six threads: beam scintillator calibration · Micromegas in beam · DAQ & trigger · simulation & trigger design · June cosmic QA · logistics & outlook

Plots marked “rendered from Claude artifact” were pulled from the interactive artifacts and rendered to image.

At a glance — what moved this weekend

Beam commissioning (July EAR2)

- Long-standing A/D coincidence deficit **traced to cabling and fixed** (run 224489); all 4 arms now equal.
- First-ever **plastic MIP** and **LIQ absolute energy scale** (Y-88 source + LIQ triples).
- Plastic HV **equalized & absolutely anchored** to the Y-88 699 keVee edge.

Micromegas in beam

- Resist-HV operating points from run-55 scan; DAQ “comb” characterised.
- Drift window resized; **zero-suppression recipe** fixed (CM subtraction mandatory).
- **762 clean 3-D micro-TPC tracks** point back at the ^{3}He source.

DAQ & trigger (n1081b + DREAM)

- Timetag “wedge” resolved: it is a **per-section rate ceiling**, not a fault.
- **Today:** full **ZS/DREAM run-config campaign** built; flash “comb” explained as fixed TCM deadtime.

Simulation & trigger design

- Final surveyed geometry; **thermal-gate trigger fully optimised** (2-arm SiPM$\times$plastic).
- Garfield HV-equivalence maps across gas mixtures.

June cosmic QA — detector-performance paper analysis matured (sub-mm resolution, spark physics, in-situ gas).

Beam commissioning — scintillator system

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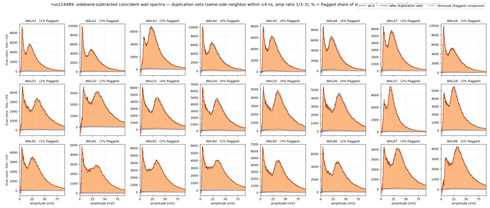
Run 224489 — the A/D deficit was cabling

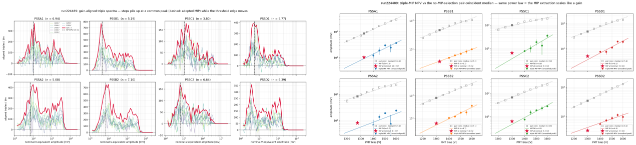
- The ± 4 ns equal-amplitude duplication band on arms A/D collapsed from **23–34%** flagged to $\leq$ 4% (B-arm control level).
- All 4 arms now show an equal wall–plastic excess (1.79/1.71/1.81/1.75 M within ± 10 ns) and clean A/D MIP bumps **with no veto**.
- First run with LIQ in readout; data quality excellent: **0/1750** bunch wall outages (vs 46% in 224404), γ-flash at 11.62 μs.
- The long-standing A/D coincidence deficit is **closed** — it was the cabling.



A/D duplication autopsy: as-is vs veto-survivor spectra now identical; every channel has a MIP bump.

First plastic MIP via LIQ triples

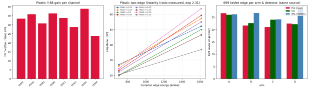
- WAL$\times$PSS$\times$LIQ triples select through-going particles $\Rightarrow$ first plastic MIP.
- MPV (linear-space) **3.3 mV (DR)** $\rightarrow$ **10.3 mV (AL)**; DR/BL MIPs sit *below* the 4.9 mV trigger $\Rightarrow$ can't fire until equalised.
- Validated by gain-alignment: scaling each HV step by $(V_{\mathrm{nom}}/V)^n$ makes a physical peak coincide while the fixed-ADC threshold edge moves.
- FIFO gain ratio $\times 1.40$ fleet geo-mean — not the naive $\times 2$.

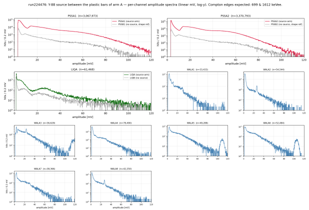


Left: gain-aligned triple-tagged spectra — MIP peak aligns, threshold edge does not. Right: triple-MIP MPV vs pair-coincident median (same power-law slope).

Y-88 source calibration — Compton edges + first LIQ energy scale

- 4 dedicated source runs (224476–79), ^{88}Y between the plastic bars.
- Both Compton edges resolved on **every** channel: 699 & 1612 keVee; measured ratio 2.40 vs ideal 2.31 ⇒ response linear to $\sim$**4%**.
- **First LIQ absolute energy scale**: 699 keVee bump on all 4 vessels, **32–37 mV/MeVee**, consistent arm-to-arm to $\sim$10%.
- All three detector types (wall / plastic / LIQ) land within 21–27 mV per arm.

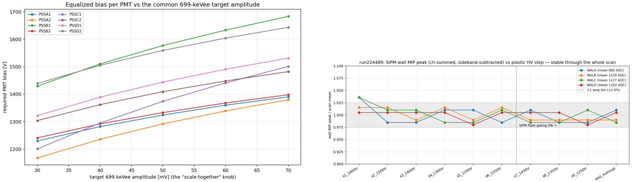




Left: plastic mV/MeVee & two-edge linearity across all detector types. Right: arm-A spectra — plastics, first LIQ Y-88 Compton bump ($\sim$26 mV), source-facing walls.

Plastic HV equalisation + nonlinearity

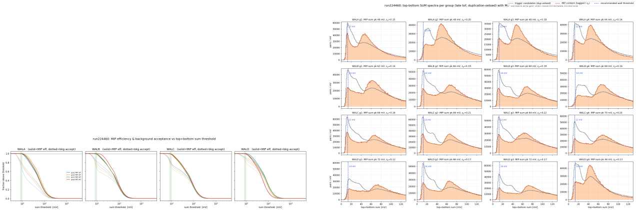
- New equalisation to fleet geo-mean coincident median (2089 ADC $= 64.2\,\mathrm{mV}$), per-PMT $n = 3.8$–7.1: ΔV from $-142\,\mathrm{V}$ (CL) to $+133\,\mathrm{V}$ (DR), **2.9**$\times$ spread.
- Absolute anchor via the Y-88 699 keVee edge $\Rightarrow$ a scale-together HV table slides the whole fleet to a common edge target.
- Nonlinearity per channel 0.78–1.13 (linear to $\sim$4%).
- **Wall MIP peak immune to plastic HV** ($\pm$2.6%, over 1200–1600 V) — validates calibration transfer.



Left: Y-88-anchored absolute plastic gain (mV/MeVee) per PMT. Right: wall MIP peak flat vs plastic HV.

SiPM wall sum-trigger threshold

- Top+bottom SUM coincidence is $\sim$**99.8% pure MIPs**; MIP sum peaks $\sim$60–80 mV (2$\times$ single-channel) on all 16 groups.
- **No MIP-vs-background valley** — the flux is MIP-dominated; amplitude alone can't reject accidentals (tighten *time* coincidence).
- Recommended sum threshold $\sim$**8 mV** ($\geq$99.5% MIP efficiency).
- Per-wall thresholds (run 224460): WALA 12 / WALB 14 / WALC 12 / WALD 14 mV.



Left: MIP efficiency vs sum threshold / accidental rate. Right: per-group sum spectra with recommended per-wall threshold lines.

Scintillator calibration — status & open flags

Frozen & in use

- LIQ absolute scale (32–37 mV/MeVee) — first time available.
- Plastic HV equalisation table (Y-88-anchored), superseding the 224466 set.
- Wall sum-trigger + per-wall thresholds.
- ADC$\rightarrow$mV $\approx$ 0.0306; plastic MIP MPV predicted 152 mV (not yet directly measured).

Open flags

- **Y-88 (35–64 mV/MeVee) vs triples-"MIP" (0.65–2.05 mV/MeV) disagree $\sim$40$\times$.** Y-88 trusted; the triples "MIP" likely sits at $\sim$130 keVee (mis-ID / low-stat) — needs a long-run recheck.
- Exact per-run plastic HV not logged in DAQsettings.
- C/D LIQ gains low (4.7 / 5.3 vs 10.4 mV on B) — LIQ HV review advised.

Micromegas in beam

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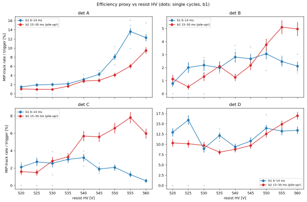
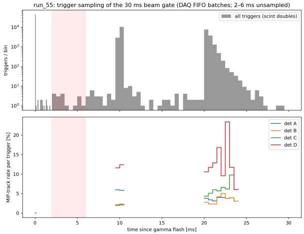
Run-55 resist-HV scan — the DAQ “comb” and operating points

- Cyclical resist scan r560→r520, all 4 dets, Ar/iso 90/10, scint-doubles, 30 ms gate; 76k triggers.
- **DAQ FIFO comb**: gate sampled only 0–0.5, ∼8–12, ∼16–28 ms — the **2–6 ms** physics target is *unsampled*.
- ^{3}He capture flood lights det D in $\gtrsim$90% of late windows.
- Saturation grows with HV; recovered by ∼8 ms at $\lesssim$550 V.

Operating points (≥3 ms): A ≥560 (try drift 700), B 550–555, C 545–550, D 540–550.

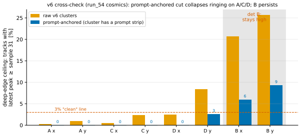


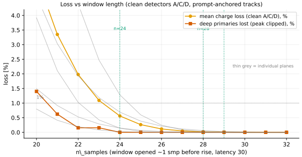


Left: b1 vs b2 MIP-track rate vs HV per det (operating point). Right: trigger comb + track rate vs time-since-flash (the 2–6 ms gap).

Drift-window sizing — latency 32 / n_samples 28

- TL;DR: latency 35 → 32, n_samples 32 → 28 now; det A drift 600 → 700 V staged; B/C/D keep 800 V.
- Earlier “slow/late B & D” was an artifact of flash/monster events + fixed FEU channel ringing (det D ceiling 8.9%→1.5%).
- Pulse footprint: rise onset peak−4, fall to baseline peak+7 (∼11 samples wide).
- **0% primaries lost down to n=24**, <0.6% mean charge (clean A/C/D).
- **The one gate is det B** (piles at the sample-31 ceiling; unresolved).

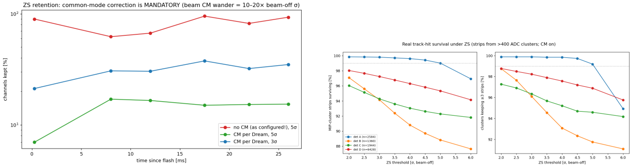




Left: prompt-anchored deep-edge check — A/C/D collapse to clean, det B stays high. Right: quantified charge/primary loss vs n_samples.

Zero-suppression — common-mode subtraction is mandatory

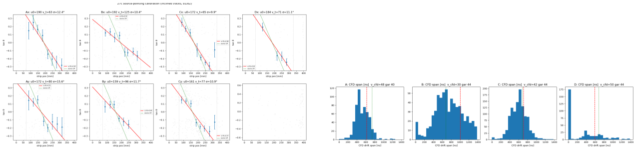
- In-beam baseline wander is **10–20× the beam-off σ** (MAD 37–96 vs 3.4–5.7 ADC).
- With no CM, even a 5σ cut keeps 60–95% of channels $\Rightarrow$ ZS useless.
- Wander is $>$99% coherent per Dream chip (64 ch); per-Dream median subtraction restores residual to **4.1 ADC** $\approx$ beam-off σ.
- **Recommendation: CM=1 + 3.5σ** $\Rightarrow$ $\sim$9.4% sample volume, $\sim$ 10× speedup — closes the comb gap.
- Strip survival @3.5σ: A 99.7 / B 92.4 / C 93.6 / D 96.8%.



Left: channel retention vs time, no-CM vs CM (the headline). Right: strip & cluster survival vs N-σ per det.

Micro-TPC tracking — clean 3-D segments point at the source

- Stringent realism gate $\Rightarrow$ **762 clean X/Y 3-D segments** (A 308, B 153, C 287, D 14); charge balance $f = 0.48$–0.50, far tighter than the shuffled null.
- **Tracks point back at the ^{3}He source** in all 8 planes; transverse alignment good to $\sim$1 cm.
- Beam-axis planes sit $\sim$3 cm low — resolved as *physics* (captures pile at the capsule bottom), independently triangulated.
- CFD drift velocities: C **42 μm/ns** matches Garfield 44; angle compression 0.62 is charge-sharing, not a v_{drift} error.



Left: $\tan\theta$ vs strip position + source fit, 8 planes. Right: CFD drift-span $\rightarrow v_{\mathrm{drift}}$ per chamber vs Garfield.

DAQ & trigger system

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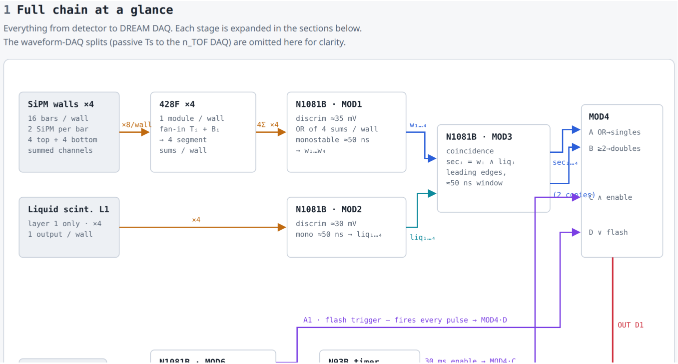
n1081b — the trigger chain

- Six CAEN N1081B logic boards (M1–M6) on a private DAQ net; WebSocket API.
- Chain: SiPM-wall OR → plastic-pair OR → **sector AND (wall × scint)** → trigger builder (singles / doubles) → DREAM.
- 20 ns coincidence window at M3; wall leg +20 ns delay.
- Discriminator hardware floor $|10\,\mathrm{mV}|$ — at the floor channels retrigger and the rate response inverts.
- Board-wedge root cause found (abandoned libwebsock 1.0.7); all access via a flock'd session wrapper.



● rendered from Claude artifact

Timetag stream & trigger-rate scans

Timetag "wedge" disproven

- A section emits **zero TT tags above a live rate ceiling** ($\sim$50 Hz streams, $\sim$800 Hz silent) — nothing is wedged; FPGA counters count at any rate.
- Consequence: walls (3–6 kHz) & liq (19–27 kHz) logged as counter totals; C/D trigger taps (tens of Hz) stream.
- Watcher **v3** = counters-first + rate gate + zero reconnect budget; offline dry-run passed, auto-start pending a hardware soak.

Post-FIFO recalibration (07-17 night)

- Plastics moved to linear fan-in/out ($\sim 2\times$ amplitude) $\Rightarrow$ all thresholds re-scanned.
- M3 timing plateau $\Rightarrow$ uniform $+20$ ns wall-leg delay; in-window fraction $0.77 \rightarrow 0.87$.
- **2-D rate scan verdict: Doubles-gated is the trigger.** Singles $\approx$1696/window (70–90$\times$ the DREAM budget); Doubles clean tail $\sim$5.6/window, $\sim$7–8 recorded/window.

The 07-15/07-17 threshold & 2-D rate scans and the flash-comb mechanism live as interactive Claude artifacts (not repo files).

Today — the ZS/DREAM run-config campaign

Stood up a live-beam **zero-suppression** run on DREAM and diagnosed the flash "comb."

Flash-comb explained

- The ~10 ms comb is a **fixed TCM BUSY→veto deadtime** (~9.6 ms at n=32), input-independent.
- TCM `event rx` 12,774 vs `tx` 1,996 ⇒ **84.4% dropped**; only ~34% live fraction at n32 (68% at n16, ~100% at n8).
- **Only NbOfSamples shortens it** (~0.32 ms/sample); ZS / IPD / SparseRd / RdClk / 10 GbE all null.

The config family (each isolates one variable)

- `zs_pulser_test` — validated Option B (Pd=0 + offline ped), tracers 100%.
- `zs_kladder` — live k-σ knee: **k8 = 6% of Raw** = safe point.
- `zs_doubles` — first ZS physics run; `zs_singles` — do findings hold at ~100× rate.
- `comb_study/readout/sparserd/vetolen/latch/cont` — swept, all null except NbOfSamples.
- `drift_scan` — overnight 2-D drift×resist (90×10 min).

Decision: open at DAQ-safe CM=1 + k8 + IPD=2, step down toward 3.5σ; write to SSD.

Simulation & trigger design

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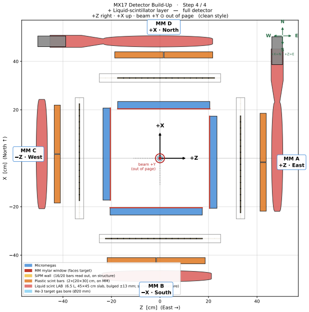
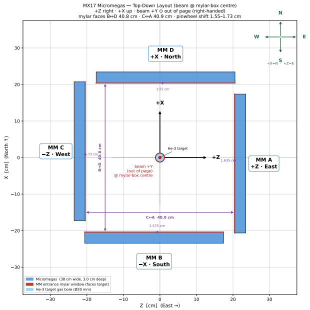
MX17 Geant4 — final surveyed geometry

- Right-handed frame, beam $= +Y$, origin at the ^{3}He target; 4 arms A/D/B/C.
- Flipped stack (2026-07-15): MM $\rightarrow$ SiPM wall $\rightarrow$ plastics $\rightarrow$ single LS layer; MM active 38 cm, 30 mm drift gap.
- LS vessel built from the `LS X17.step` CAD: 45 cm slab, 6.5 L LAB, PMT vertical on B/C, horizontal on A/D.
- Pinwheel tangential shifts corrected (earlier values were $2\times$ too large).
- Open item: sim outputs stale — acceptance must be re-run.



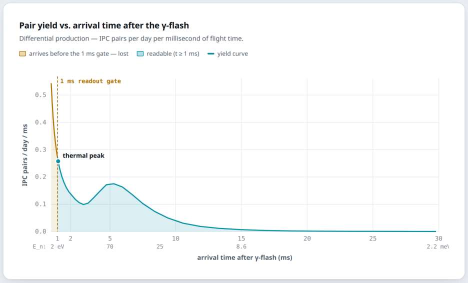


Left: full detector-stack buildup MM→SiPM→plastics→LS. Right: 4-arm MM layout, top-down.

Thermal trigger — the physics sets the window

- Below $\sim$2 eV the pair yield is fixed by the beam's thermal Maxwellian — you can only decide **how early you dare to read out.**
- A $>$1 ms readout gate $\Leftrightarrow E_n < 1.99$ eV at 19.5 m; **44% of the thermal beam arrives before 1 ms and is lost** to the flash + readout deadtime.
- 99.9% of usable arrivals land in the 1–30 ms window.
- Background-characterisation regime, not discovery — the physics case still lives at 0.1–10 MeV.

READABLE IPC / DAY	READABLE X17 / DAY	PEAK ARRIVAL	LOST BEFORE 1 MS	CONTAINED 1–30 MS
1.22	0.031	≈5 ms	44%	99.9%
arriving after 1 ms (E < 1.99 eV)	≈ 1 X17 every 33 days	E_n ≈ 70 meV (thermal bump)	0.96 IPC/day at 2 eV–1 keV	last arrivals by ~45 ms (1 meV)



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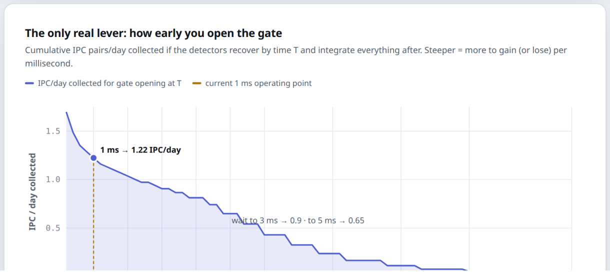
Thermal trigger — how early to open the gate

- The one real lever: cumulative IPC/day collected if the detectors recover by time T.
- Current 1 ms operating point $\rightarrow$ **1.22 IPC/day**; 3 ms drops to $\sim$0.9, 5 ms to 0.65.
- MIP calibration: **1 MIP = 458 keV (SiPM), 4.33 MeV (plastic)**, validated against the beam Landau $\Rightarrow$ thresholds transfer directly to hardware.
- Folding $\sim$3–5% double-trigger acceptance $\rightarrow$ $\sim$0.05 detected IPC/day ($\sim$1–1.5 per 25-day run).



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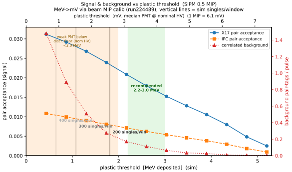
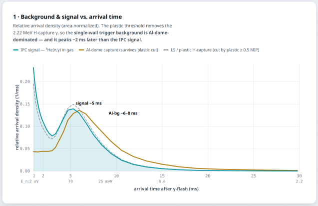
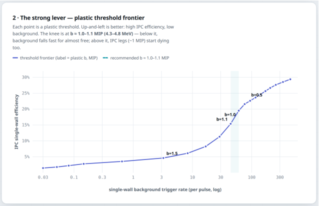


Bottom (Geant4): signal + background vs plastic threshold with DAQ ceiling lines.

Trigger — single-wall background & the plastic-threshold lever

- Two knobs protect IPC events: **when you read out** and **the plastic threshold**.
- The threshold is the strong lever (rate drops by orders of magnitude). The surviving background is Al-dome 7.72 MeV γ (H-capture), arriving *later* than the signal.
- Key result: **plastic threshold does not bias IPC vs X17** (ratio flat $\sim$0.35) — a free rate/purity knob costing only total statistics.
- Chosen point: full 2-arm SiPM$\times$plastic, SiPM $\geq$0.5 MIP, plastic $b \approx$ 2.6–3.0 MeV.





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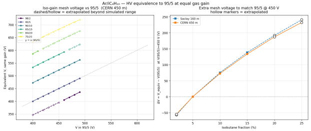
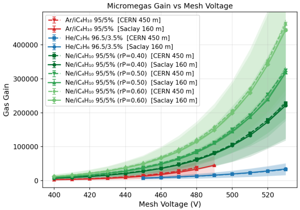
Left: background & signal vs arrival time. Right: the plastic-threshold frontier (IPC efficiency vs single-wall rate).

Garfield gas / HV equivalence

- Maps every Ar/iC_4H_{10} mixture's mesh HV to the 95/5 reference at equal simulated gain (Garfield++/Magboltz).
- Reference 95/5 gain over 400–490 V rises $\sim \times 17$ **per 90 V**.
- More quencher $\Rightarrow$ higher operating voltage: at the 450 V (95/5) point, 90/10 needs 525 V, 85/15 589 V, 80/20 642 V, 75/25 691 V.
- Gain model $\ln G$ quadratic in V, $R^2 \geq 0.997$; two site pressures (Saclay, CERN).





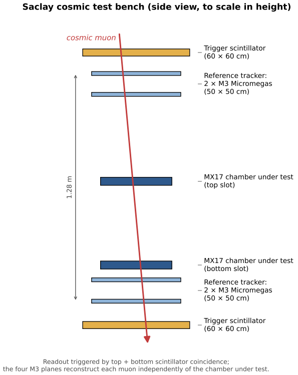
Left: HV-equivalence map across mixtures. Right: simulated gas gain vs mesh voltage.

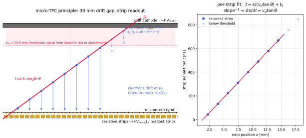
June cosmic QA — detector-performance paper

Section 5 of 6

Cosmic bench — the micro-TPC principle

- Saclay bench: 5 resistive-strip micro-TPC Micromegas (det 2/3/4/6/7) vs an independent M3 telescope reference.
- 512 X + 512 Y strips, **0.78 mm pitch**, $\sim 40 \times 40\,\mathrm{cm}^2$; 30 mm drift gap run as a micro-TPC (drift time $=$ depth).
- One 30 mm gap is a self-contained 3-D tracker: sub-pitch position, $\sim 1.7°$ angle, ~ 1 mm-equiv timing.



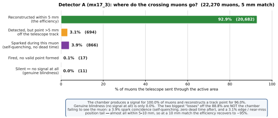


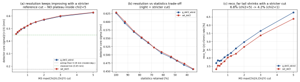
Left: bench — scintillator trigger + 4 M3 planes sandwiching 2 MX17 slots.

Right: micro-TPC principle.

Fleet efficiency & the M3 reference recipe

- Residuals were **reference-limited, not chamber-limited**: σ keeps falling with a tighter M3 χ^2 cut with no plateau.
- Recommended recipe $\chi^2 < 1.0$ & NClus$=4$: core σ 0.63 $\to$**0.47 mm**, efficiency $\to$92.9% (det3).
- Fleet: det3 92.9% / det2 91.3% healthy; det6 57.8% & det7 43.1% spark-limited; det4 20.7% gain-limited.
- Y-plane passivation common to all 5 (17.9–20.4 mm) — a construction feature.

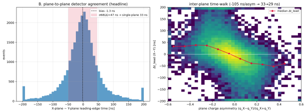


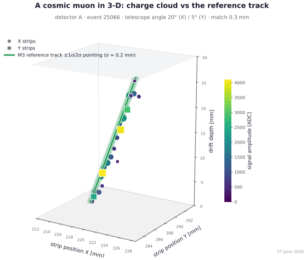


Left: det3 loss budget (92.9% reco, 3.9% spark). Right: σ / reco-far vs M3 χ^2 cut — no plateau.

Resolution & a 3-D event display

- Geometry-free plane-to-plane timing: **33 ns per plane ≈ 1 mm of drift**.
- Deconvolved DUT intrinsic position $\sigma \approx 0.40$ mm; M3 self-resolution measured directly (X planes 0.41 mm).
- XY charge balance $f \approx 0.5$, position- & angle-independent (Pearson $r = 0.881$) — the pixel routing works.
- 3-D displays show the **independent M3 track threading the drift charge cloud** (match 0.1–0.3 mm).





Left: 33 ns (≈1 mm) from two orthogonal planes. Right: one muon in 3-D, reference track through the charge cloud.

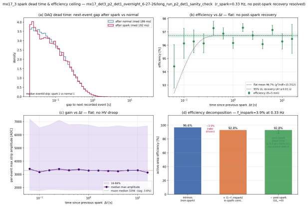
Spark physics & in-situ gas

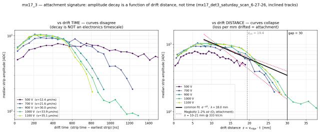
Sparks

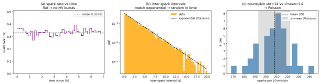
- Poisson/random in time (flat 0.33 Hz), muon-*induced* (4.9% on a crossing vs 1.2% on a miss), edge-seeded.
- **No measurable post-spark dead time** — efficiency & gain flat vs time-since-spark; resistive self-quenching seen in the raw waveforms.

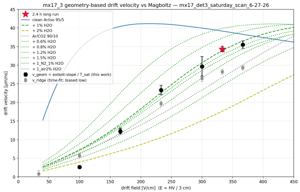
Gas, in situ

- $v_{\mathrm{drift}} = 34 \pm 1.5\ \mu$m/ns; tracks the Magboltz "Ar/iso + $\sim$1% H_2O" family.
- Amplitude decays with drift *distance* ($\lambda \approx 18$ mm) $\Rightarrow$ O_2/H_2O attachment.







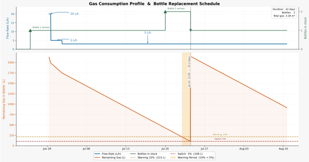


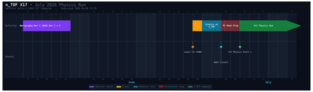
Logistics, safety & outlook

Section 6 of 6

Logistics — gas budget & campaign timeline

- Gas model: 2150 L bottle; steady $\sim$3 L/h $\Rightarrow$ **one bottle $\sim$30 days**, $\sim$3280 L for the run.
- Swap thresholds: warn at 10% (215 L), switch at 5%; deliveries scheduled.
- July timeline: build 8–26 Jun, EAR2 install 29 Jun, X17 physics from 1 Jul.





Left: gas consumption / bottle depletion with warning bands. Right: July build + EAR2 campaign timeline.

DAQ throughput — the ZS-IPD safety case

- Deadtime is set almost entirely by the FEU inter-packet delay ($\tau \approx 32\,\mu$s$\times$IPD $+ \sim$140 μs SCA), *not* the ZS threshold.
- Lowering IPD lifts the ceiling: **18 events/30 ms window at IPD=100 $\rightarrow$ 226 at IPD=1–2** — but exposes host-buffer overflow, 1 GbE saturation, and FEU event-size truncation.
- Losses are **never silent** (eventId gaps) + tracer channels (8/FEU) + `zs_integrity.py` detect ZS loss.
- Beam-validated: **k8 + IPD=2 records 165 events/window (9.2$\times$ current) at 0.9% tagged loss**; a $\sim$300 EUR 10 GbE upgrade reaches $\sim$220–260/window.
- This defines the "$\sim$200 events/window" DAQ ceiling that the thermal trigger budgets against.

Open issues & next steps

Open issues

- **Y-88 vs triples-"MIP" $\sim$40$\times$ discrepancy** — resolve with a long-run recheck; per-run plastic HV not logged.
- **Det B drift** piles at the sample ceiling (real slow drift vs residual coherent noise — unresolved).
- **2–6 ms window never sampled** in run-55 — ZS + shorter n_samples needed to reach it.
- Sim acceptance stale — must re-run on the new geometry.

Next steps

- Launch the ZS/DREAM run (CM=1, k8, IPD=2 $\rightarrow$ 3.5σ); include a no-ZS reference subrun.
- Re-scan the MM operating points with the comb closed (measure 1–8 ms).
- Soak-test the TT watcher v3, then enable auto-start.
- Apply the Y-88 plastic HV table; measure the plastic MIP directly at nominal.
- Re-run Geant4 acceptance $\rightarrow$ finalise the trigger operating point.

Deliverables — reports, decks & artifacts

Compiled reports & decks

- Beam QA: `run224489`, `y88`, `hv_scan`, `mip` reports; drift-window deck.
- June cosmic: det3-recofar (`report_v2`), M3 self-resolution, engineer package (00–73 figures + event displays).
- Sim: thermal-trigger deck; ZS-IPD DAQ safety report.

Interactive Claude artifacts

- *Rendered into this deck:* thermal X17/IPC yield; single-wall trigger background; n1081b trigger logic.
- *Available (not on disk):* DREAM flash-comb anatomy; scint recalibration; 07-15/17 threshold & 2-D rate scans.

Compiled from `~/PycharmProjects/nTof_x17{,_DAQ}`, `CLionProjects/MX17_Full_Geant`, and the 18–19 Jul weekend collection.

Thank you

Questions, and detail on any thread, welcome.